



SIMATS
ENGINEERING



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**GREEN BANKING AND SUSTAINABLE
LOAN MANAGEMNET SYSTEM**
A CAPSTONE PROJECT
**DSA0101 OBJECT ORIENTED PROGRAMMING WITH
C++**

FACULTY

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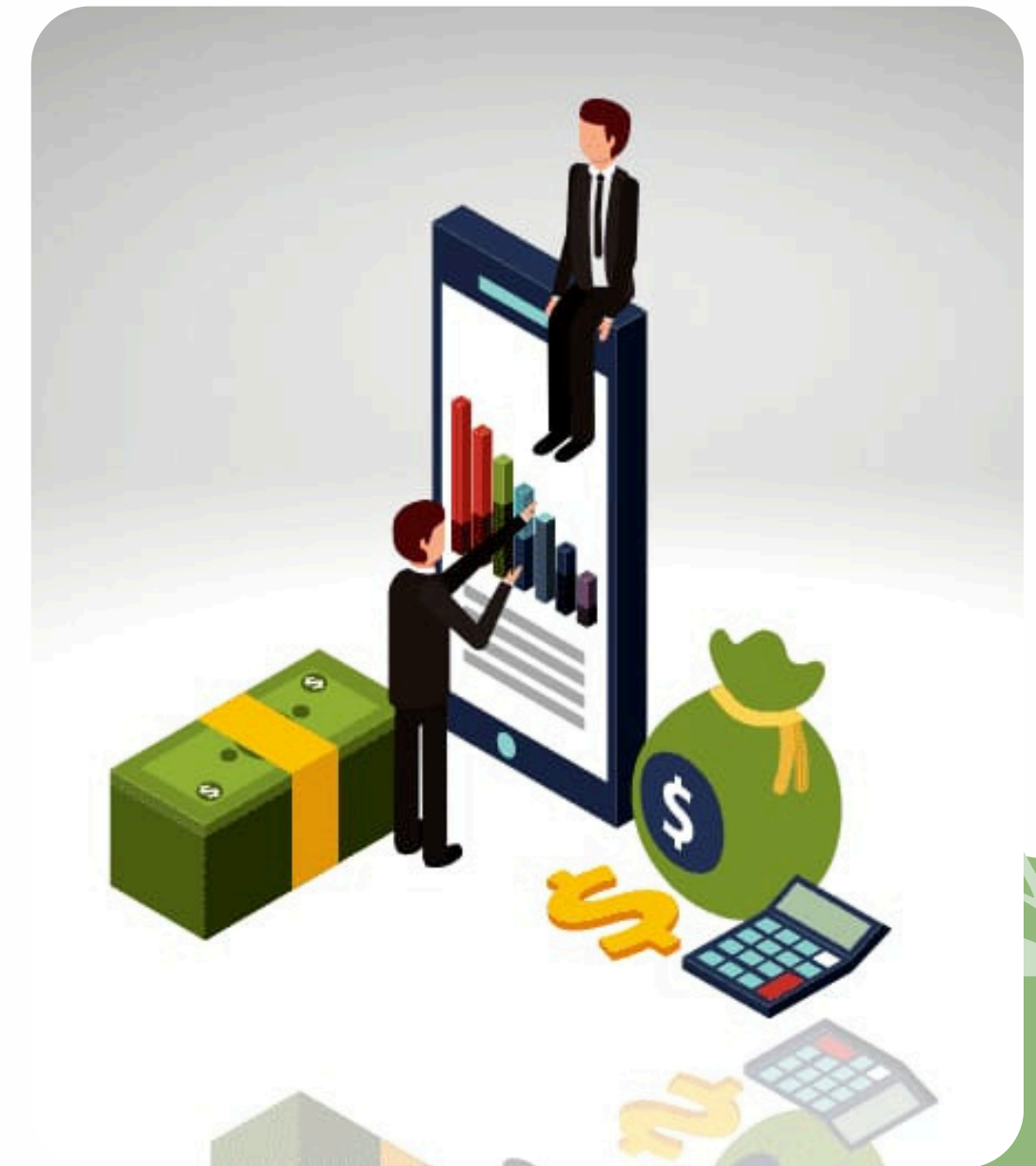
ABSTRACT

Green banking has become an essential strategy for promoting sustainable economic growth while protecting the environment. It encourages banks to adopt environmentally responsible practices in their operations and lending decisions. Banks play a vital role in supporting sustainable development by financing projects that reduce pollution, improve energy efficiency, and promote the use of renewable resources. Green banking also helps financial institutions assess environmental risks before approving loans, ensuring that businesses follow environmental regulations and sustainable practices.



INTRODUCTION

- Green Banking promotes eco-friendly and paperless banking
- Encourages digital banking services and sustainable financial practices
- Supports green projects such as renewable energy and electric vehicles
- Our system is developed using C++
- Manages customer information and sustainable loan applications



LIRERATURE SURVEY

S.No	Author & Year	Methods	Pros	Cons
1	Sahoo & Nayak (2021)	Analysed sustainable finance and credit risk assessment for green loans.	Provides risk assessment factors for sustainable loans.	No automated scoring or approval process.
2	Singh & Sharma (2022)	Studied ESG criteria integration in banking and its effect on loan performance.	Shows positive impact of ESG-based lending on loan performance.	Complex ESG data collection for small borrowers.
3	Patel & Patel (2023)	Proposed a digital model for green loan management in commercial banks.	Demonstrates digital workflow for green loan tracking.	Does not include sustainability awareness tracking.
4	Rathi & Mishra (2024)	Reviewed AI and machine learning techniques for sustainable finance.	AI improves accuracy in credit and sustainability risk prediction.	Requires large datasets and infrastructure.
5	Verma & Gupta (2024)	Developed a framework for green banking with real-time sustainability monitoring.	Real-time monitoring improves accountability and transparency.	Higher implementation cost and complexity.

OBJECTIVES

- To design a loan management system using Object-Oriented Programming in C++
- To integrate an eco-score mechanism that rates the sustainability of loan projects
- To offer reduced interest rates for environmentally friendly loans
- To ensure loans are approved based on both eco-score and creditworthiness
- To maintain customer records and track loan repayments through separate modules
- To promote green banking practices through a functional, modular software system

ISSUES IN THIS EXISITING MODEL

1

Banks focus only on financial indicators (profit, collateral, repayment history) and ignore environmental risks

2

Traditional loan processing relies heavily on paper, increasing paper waste.

3

Lack of awareness about green banking services among customers.

4

No standardized framework for evaluating green loans and sustainability.

PROPOSED SOLUTION

1

Develop a Green Banking and Sustainable Loan Management System with a digital, paperless workflow.

3

Use AI-based ESG (environmental social governance) verification and standardized metrics

2

Develop an integrated Green Banking and Green Financing model to achieve SDGs (sustainable development goal).

4

Use paperless online loan application and approval processes

1

**MODULE 1 -
Customer Management Module**

2

**MODULE 2 -
Green Loan Application Module**

3

**MODULE 3 -
Eco-Score Assessment Module**

4

**MODULE 4 -
EMI Calculation Module**

5

**MODULE 5 -
Loan Approval Module**

6

**MODULE 6 -
Transaction & Report Module**



MODULE 1 -

Customer Management Module

- Registers new customers
- Stores basic KYC details – Customer ID, Name, Email
- Maintains customer account balance
- Allows balance to be checked and updated
- Keeps identity and account data separate from green-loan logic
- A customer can apply for multiple loans
- Uses custId as the reference for other modules
- Acts as the foundation of the entire system

Digital Loan Management System



SAMPLE CODE

```
1 class Customer {
2 public:
3     int custId;
4     string name;
5     string email;
6     double accountBalance;
7
8     Customer(int id, string n, string e, double bal) {
9         custId = id; name = n; email = e; accountBalance = bal;
10    }
11
12    void display() {
13        cout << custId << " | " << name << " | " << email
14            << " | Balance: " << accountBalance << "\n";
15    }
16 };
```

Output - Module 1: Customer Management

```
1 | Anamika | anamika@mail.com | Balance: 5000
```

```
Process finished with exit code 0
```

MODULE 2 - Green Loan Application Module

- Captures eco-friendly loan requests from customers
- Stores Loan ID, Customer ID, amount, and loan purpose
- Receives and stores the Eco Score from the Eco-Score Module
- Initially sets every loan status as “Unapproved”
- Sends the loan for processing to the Approval Module



SAMPLE CODE

```
1 class GreenLoan {
2 public:
3     int loanId;
4     int custId;
5     double amount;
6     string purpose;
7     int ecoScore;    // rating 0-100 based on project sustainability
8     bool approved;
9
10    GreenLoan(int lid, int cid, double amt, string p, int score) {
11        loanId = lid; custId = cid; amount = amt;
12        purpose = p; ecoScore = score; approved = false;
13    }
14
15    void display() {
16        cout << loanId << " | Customer " << custId << " | " << purpose
17             << " | Amount: " << amount << " | EcoScore: " << ecoScore << "\n";
18    }
19 };
```

Output - Module 2: Green Loan Application

```
101 | Customer 1 | Solar Installation
Amount: 200000 | EcoScore: 90

Process finished with exit code 0
```

MODULE 3 - Eco-Score Assessment Module

- Evaluates how sustainable a project is
- Accepts either a project type or an energy-saving value
- Assigns a numeric score between 0–100
- Kept separate so scoring rules can be updated independently
- Feeds directly into approval and EMI decisions



SAMPLE CODE

```
1 class EcoScore {
2 public:
3     int calculateScore(string projectType) {
4         if (projectType == "Solar") return 90;
5         else if (projectType == "EV") return 80;
6         else if (projectType == "Energy Efficient Home") return 75;
7         else return 50;
8     }
9
10    int calculateScore(double energySavedKwh) {
11        if (energySavedKwh > 1000) return 95;
12        else if (energySavedKwh > 500) return 80;
13        else return 60;
14    }
15 };
```

Output - Module 3: Eco-Score Assessment

Project Type: Solar
Eco Score: 90

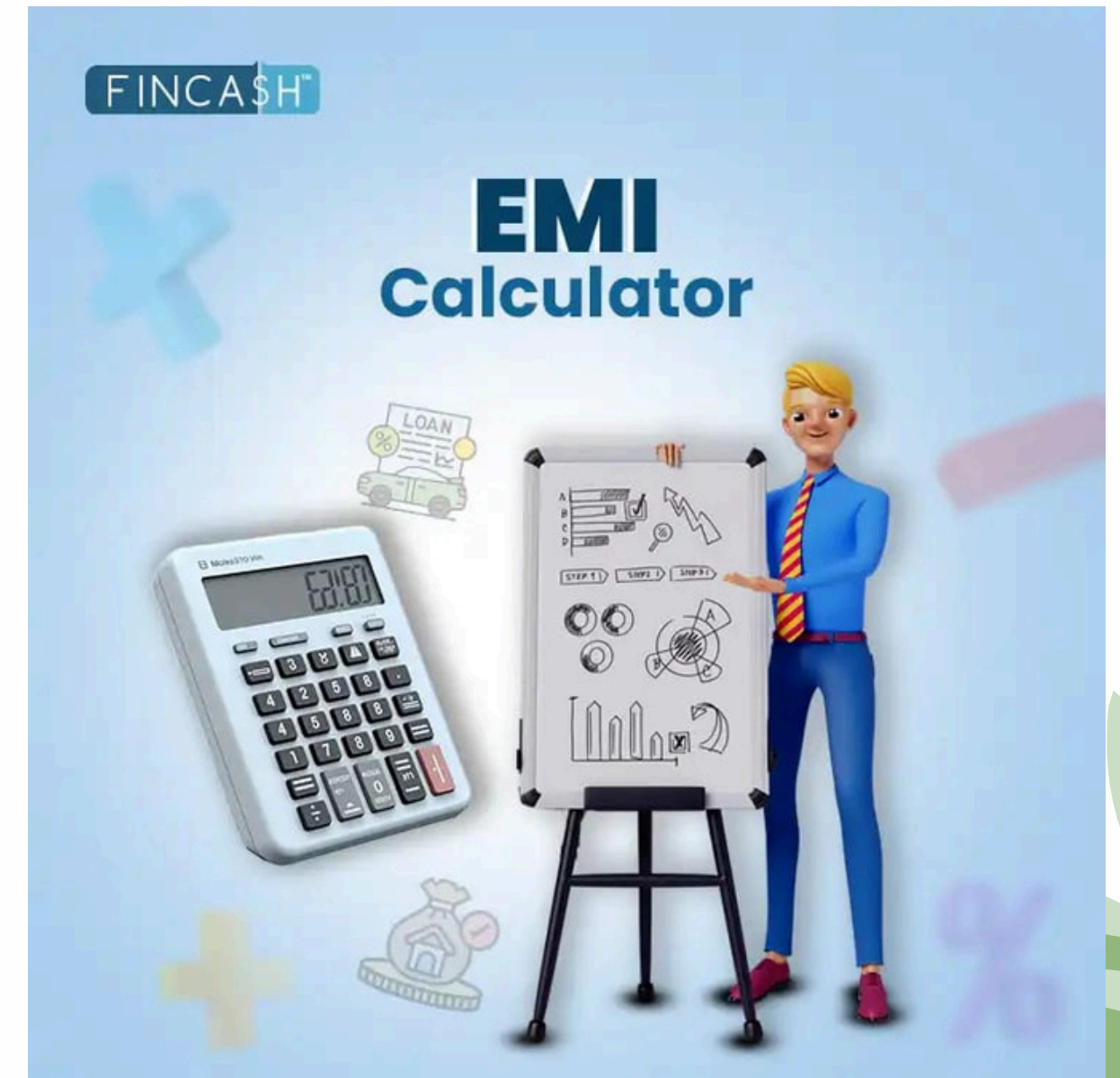
Project Type: EV
Eco Score: 80

Energy Saved: 1200 kWh
Eco Score: 95

Process finished with exit code 0

MODULE 4 - EMI Calculation Module

- Computes monthly installment from loan amount, rate, and tenure
- Applies a 0.5% interest discount for high eco scores (>80)
- Uses the standard compound-interest EMI formula
- Converts annual rate into monthly rate internally
- Rewards sustainable loans with lower repayments



SAMPLE CODE

```
1 #include <cmath>
2
3 class EMICalculator {
4 public:
5     double calculateEMI(double principal, double annualRate, int months, int ecoScore) {
6         double rate = annualRate;
7         if (ecoScore > 80) rate -= 0.5;    // green discount
8
9         double monthlyRate = (rate / 12) / 100;
10        double emi = (principal * monthlyRate * pow(1 + monthlyRate, months)) /
11                    (pow(1 + monthlyRate, months) - 1);
12        return emi;
13    }
14};
```

Output - Module 4: EMI Calculation

Test 1: Principal=200000, Rate=9.0, Months=24, EcoScore=90
Monthly EMI: 9091.13

Test 2: Principal=200000, Rate=9.0, Months=24, EcoScore=60
Monthly EMI: 9136.95

Process finished with exit code 0

MODULE 5 - Loan Approval Module

- Checks eco score against a minimum threshold (≥ 70)
- Checks credit score against a minimum threshold (≥ 650)
- Approves only if both conditions are met
- Rejects and shows a reason if either fails
- Acts as the decision gateway of the whole system



SAMPLE CODE

```
1 class LoanApproval {
2 public:
3     bool approveLoan(int ecoScore, double creditScore) {
4
5         if (ecoScore >= 70 && creditScore >= 650) {
6
7             cout << "Loan Approved with Green Benefits!\n";
8             return true;
9
10        } else {
11
12            cout << "Loan Rejected.\n";
13            return false;
14
15        }
16    }
17};
```

Output - Module 5: Loan Approval

Test 1: EcoScore=90, CreditScore=700
Loan Approved with Green Benefits!

Test 2: EcoScore=60, CreditScore=600
Loan Rejected.

Process finished with exit code 0

Transaction & Report Module

- Records every EMI/repayment made by the customer
- Stores payments in a running list (vector) tied to the loan
- Each new payment is appended without overwriting previous entries
- Calculates the total amount repaid so far by summing all payments
- Displays full repayment history, listing each individual payment
- Helps the customer track how much of the loan is still pending
- Acts as the final stage in the loan lifecycle — apply → score → approve → repay → report



SAMPLE CODE

```
1 #include <vector>
2
3 class TransactionManager {
4 private:
5     vector<double> payments;
6
7 public:
8     void recordPayment(double amt) {
9         payments.push_back(amt);
10    }
11
12    void generateReport() {
13
14        double total = 0;
15        cout << "----- Repayment Report -----\\n";
16
17        for (double p : payments) {
18            cout << "Payment: " << p << "\\n";
19            total += p;
20        }
21
22        cout << "Total Repaid: " << total << "\\n";
23    }
24};
```



Output - Module 6: Transaction & Report

----- Repayment Report -----

Payment: 9091.13

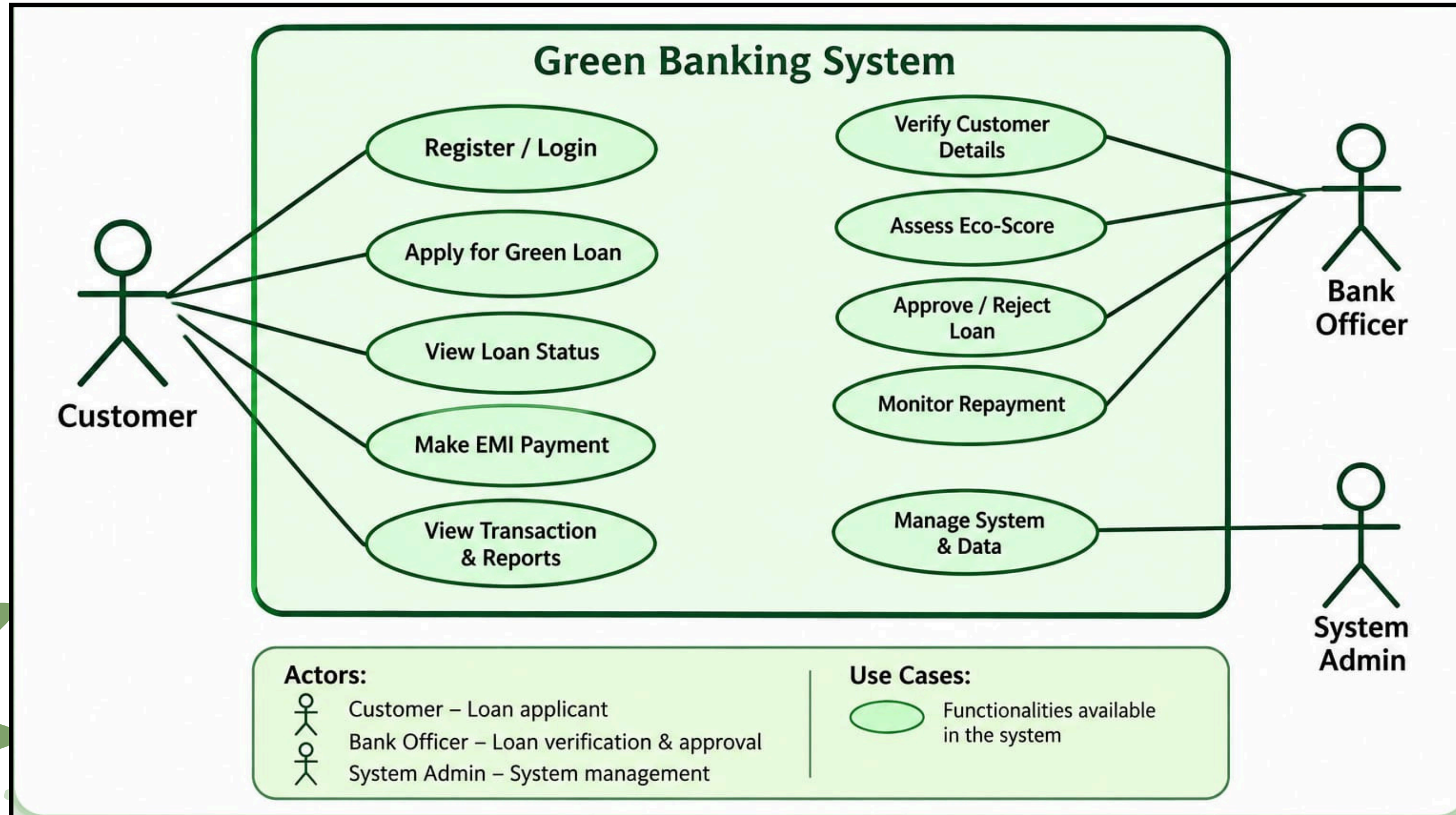
Payment: 9091.13

Payment: 9091.13

Total Repaid: 27273.4

Process finished with exit code 0

USE CASE DIAGRAM



SYSTEM WORKFLOW

**START → Customer Registration → Enter
Green Loan Application → Eco-Score
Assessment → Calculate EMI & Interest
Discount → Check Eligibility (Eco-Score ≥ 70
& Credit Score ≥ 650) → Loan Approved /
Rejected → If Approved → EMI Repayment
→ Record Transactions → Generate
Repayment Report → END**

TECHNOLOGY AND CONCEPTS USED

Technology Used:

- **C++ – Main programming language used to develop the system**
- **Object-Oriented Programming (OOP) – Used to design the system in separate modules.**
- **Vector – Used to store repayment/transaction records**

Concepts Used:

- **OOP – Organizes the system using objects**
- **Classes & Objects – Manages customer and loan data**
- **Encapsulation – Protects and organizes data.**
- **Modular Design – Divides the system into separate modules**
- **Decision Making – Checks eligibility for loan approval**

ADVANTAGES

1



PAPERLESS BANKING

Reduces paper usage and minimizes environmental waste by digitizing the loan process.

2



ECO-FRIENDLY LOANS

Promotes and supports sustainable and green projects that help protect the environment.

3



ECO-SCORE SYSTEM

Rates projects based on their environmental impact and encourages better sustainability practices.

4



LOWER INTEREST

Projects with higher eco-score receive interest discounts and better loan benefits.

5



EASY REPAYMENT TRACKING

Maintains repayment history and generates reports for easy tracking and better management.

FUTURE SCOPE

- **AI-based Eco-Score – Improve the accuracy of environmental assessment using AI**
- **Mobile Application – Provide easy access to loans, EMI payments, and account tracking**
- **Blockchain Integration – Ensure secure and transparent transaction records**
- **Real-time Monitoring – Track environmental activities and update the Eco-Score regularly**
- **Integration with Green Projects – Connect the system with renewable energy, electric vehicles, and other sustainable projects**

CONCLUSION

- Applied core OOP concepts to build a real financial system
- Eco-score rewards green projects with lower interest
- Modular design keeps the system maintainable
- Balances sustainability with credit risk management
- Scope to add carbon tracking and mobile access

**GREEN
BANKING**



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- **RBI. (2022). Guidelines on green deposits and climate risk management. Reserve Bank of India Bulletin.**
- **Bose, S., & Khan, H. (2020). Green banking and environmental sustainability in Indian banks: A critical analysis. Indian Journal of Finance, 14(4), 21–30.**



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Thank You



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